

Fake News Detection Using TF-IDF and Logistic Regression

Max Mendez

Section 1

jam1116

Repository: <https://github.com/maxmendez13/rutgers-ds-spring-26-final-project>

Abstract

The spread of misinformation through online news platforms has become a significant societal challenge, increasing the need for automated systems capable of identifying fake news articles accurately and efficiently. This project presents a machine learning-based fake news detection system using Natural Language Processing (NLP) techniques. A combined dataset of real and fake news articles, provided by Kaggle, was preprocessed through text cleaning and normalization procedures, including lowercasing, punctuation removal, and stop-word filtering. The textual data was transformed into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization. A Logistic Regression classifier was then trained to distinguish between fake and real news content. The model was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Experimental results demonstrated that the proposed approach achieved strong classification performance, showing that TF-IDF features combined with Logistic Regression provide an effective and computationally efficient solution for fake news detection. These findings highlight the potential of machine learning techniques in combating misinformation in digital media environments.

1 Introduction

The rapid growth of digital media and online news platforms has transformed how people access and consume information. While these technologies provide instant access to global events, they have also contributed to the widespread dissemination of misinformation and fake news. Fake news refers to intentionally misleading or fabricated information presented as legitimate journalism. The spread of false information through social media and online publishing platforms can influence public opinion, damage trust in reliable news sources, and affect important societal processes such as elections, public health decisions, and political discourse. As the volume of online content continues to increase, manually verifying the credibility of news articles has

become increasingly difficult, creating a strong need for automated fake news detection systems.

This project was motivated by a personal interest in journalism and reliable news reporting. Many individuals, including the author, have encountered misleading online information that appeared credible at first glance. The ability to automatically identify misinformation could help users navigate the modern media landscape more effectively and reduce the influence of deceptive content.

This project addresses the problem of automatically classifying news articles as either fake or real using machine learning and Natural Language Processing techniques. The proposed system uses TF-IDF vectorization to transform textual news data into numerical feature representations and applies a Logistic Regression classifier for binary classification. TF-IDF was selected because it effectively captures word importance within documents while reducing the impact of frequently occurring terms. Logistic Regression was chosen due to its efficiency, interpretability, and strong performance in text classification tasks.

The primary contributions of this project include the development of a complete fake news detection pipeline involving preprocessing, feature extraction, model training, and evaluation. The project further demonstrates that relatively lightweight and computationally efficient machine learning methods can achieve strong performance in misinformation detection tasks.

2 Related Work

Fake news detection has become an important research area due to the rapid spread of misinformation on digital platforms and social media. Many studies have applied machine learning and Natural Language Processing techniques to automatically classify news articles as real or fake. Traditional predictive modeling approaches commonly use text vectorization methods such as Bag-of-Words and TF-IDF combined with classifiers including Logistic Regression, Naive Bayes, and Support Vector Machines [1]. Prior research has shown that TF-IDF and Logistic Regression are effective because they can efficiently analyze high-dimensional textual data while maintaining strong classification performance [2].

More recent research has explored hybrid and ensemble frameworks that combine multiple algorithms or integrate additional contextual information such as social media behavior, source credibility, and user interaction patterns. Ensemble approaches using TF-IDF with classifiers such as Random Forests, Bagging, and boosting methods have demonstrated improved robustness and classification accuracy in fake news detection tasks [3].

Compared to more computationally intensive deep learning and hybrid approaches, this project focuses on a lightweight and interpretable framework using TF-IDF vectorization and Logistic Regression classification. The results demonstrate that traditional machine learning methods remain highly effective for fake news

detection while requiring fewer computational resources and simpler implementation.

3 Methodology

This project utilized the Fake News Detection dataset obtained from Kaggle, which consisted of two comma-separated value (CSV) files: *Fake.csv* and *True.csv*. The *Fake.csv* dataset contained fake news articles, while the *True.csv* dataset contained legitimate news articles. The title and text fields were selected as the primary features because they contained the most relevant linguistic information for classification.

To prepare the data for supervised learning, binary labels were assigned to each article. Fake news articles were mapped to the value 1, while real news articles were mapped to the value 0. The datasets were then concatenated and randomly shuffled to improve model generalization.

Several preprocessing techniques were applied before training the model. The title and article body were merged into a single feature called *content*. The text was converted to lowercase, punctuation was removed, and stop words were filtered to reduce noise and standardize formatting across documents.

The cleaned textual data was transformed into numerical representations using TF-IDF vectorization. TF-IDF measures the importance of words within documents relative to the overall dataset, allowing the model to focus on more informative terms. The vectorizer was configured with English stop-word removal and `max_df=0.7` to ignore extremely common terms.

After preprocessing, the dataset was divided into training and testing subsets using an 80/20 train-test split. A Logistic Regression classifier with `max_iter=1000` was then trained on the TF-IDF feature vectors. Logistic Regression was selected because it performs efficiently on high-dimensional sparse text data and provides interpretable classification behavior.

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics provided a comprehensive assessment of the effectiveness of the proposed fake news detection system.

4 Experiments and Results

Experiments were conducted using the Kaggle Fake News Detection dataset to evaluate the effectiveness of the proposed classification system. After preprocessing and TF-IDF vectorization, the Logistic Regression classifier was trained and evaluated using an 80/20 train-test split.

Accuracy: 0.9856347438752784

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.99	0.98	4284
1	0.99	0.99	0.99	4696
accuracy			0.99	8980
macro avg	0.99	0.99	0.99	8980
weighted avg	0.99	0.99	0.99	8980

Figure 1: Classification Report

The model achieved strong classification performance across multiple evaluation metrics, demonstrating its ability to accurately distinguish between fake and real news articles. Accuracy served as the primary evaluation metric, while precision and recall measured the model’s effectiveness in identifying fake news articles and minimizing classification errors. Confusion matrix analysis further showed that the majority of articles were correctly classified across both categories.

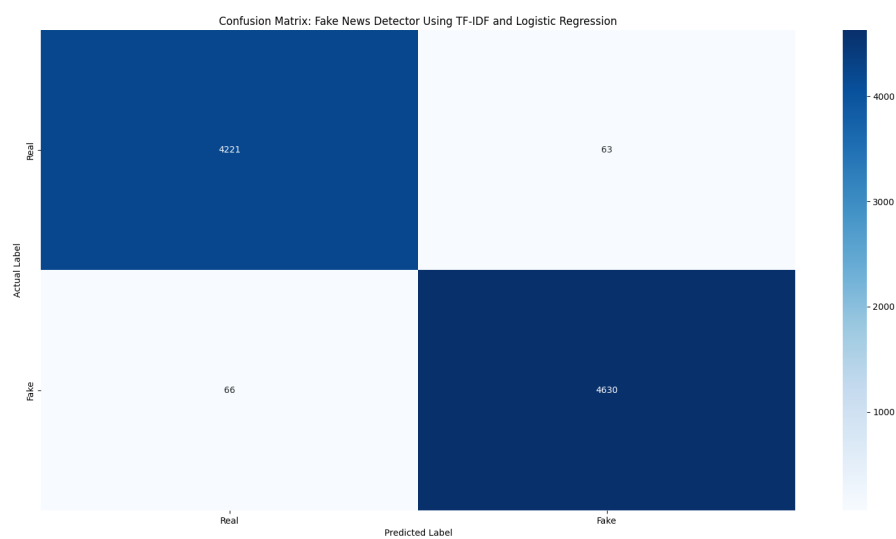
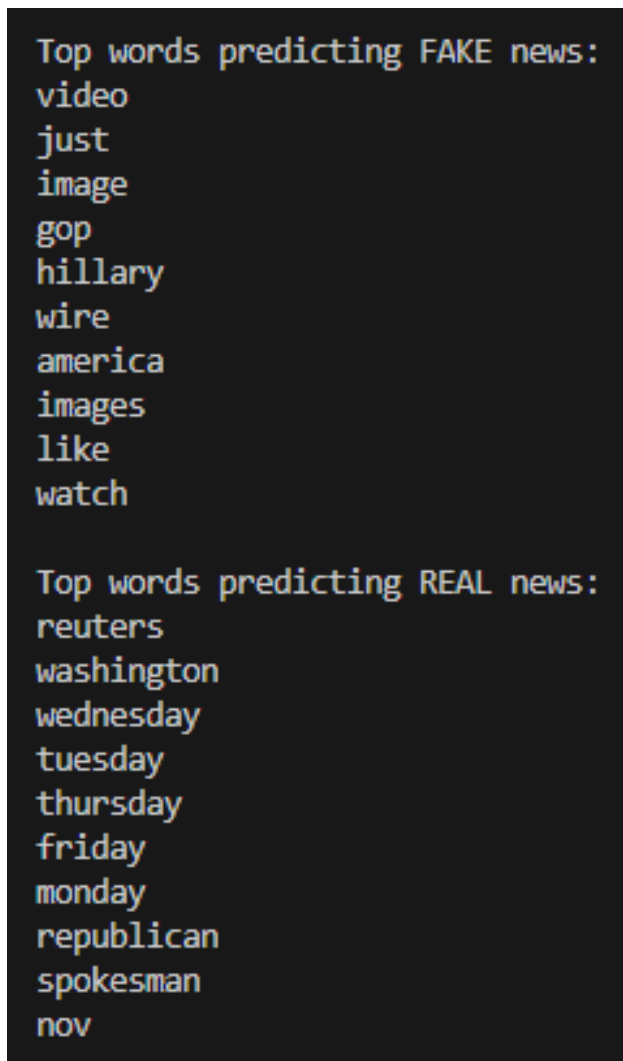


Figure 2: Confusion Matrix Heatmap

Additional analysis was performed using feature importance extracted from the Logistic Regression coefficients. Terms with high positive coefficients were more strongly associated with fake news predictions, while other terms were associated with real news articles. Visualization techniques such as confusion ma-

trix heatmaps and feature importance plots were generated to improve interpretability and support the experimental findings.



```
Top words predicting FAKE news:
video
just
image
gop
hillary
wire
america
images
like
watch

Top words predicting REAL news:
reuters
washington
wednesday
tuesday
thursday
friday
monday
republican
spokesman
nov
```

Figure 3: Top Words Indicating Real or Fake News

Overall, the experiments demonstrate that TF-IDF vectorization combined with Logistic Regression provides an efficient and effective approach for fake news detection while maintaining relatively low computational complexity.

5 Conclusion

This project demonstrated that combining TF-IDF vectorization with Logistic Regression provides an effective and computationally efficient approach for fake news detection. The model successfully classified news

articles as fake or real with strong performance across multiple evaluation metrics. The findings show that traditional machine learning and Natural Language Processing techniques remain highly capable for text classification tasks involving misinformation detection.

Despite the strong performance, several limitations exist within the proposed approach. The model relied only on textual content and did not incorporate contextual information such as publisher credibility, user behavior, or social media engagement. Additionally, traditional machine learning models may struggle to capture deeper semantic relationships within language compared to modern deep learning approaches.

Future work could improve the system by incorporating transformer-based models such as BERT, integrating social media metadata, and experimenting with ensemble learning techniques. Expanding the dataset and applying real-time misinformation detection methods could further enhance the practical applicability of the model.

References

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